Exploring Candidate RNN Perturbations for Psilocybin-Related Working-Memory Effects

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METHODS

Fixation-gated circular delayed-response task

Each circular-task trial comprised pre-cue fixation, a circular cue, a silent memory delay, and a response period. Stimulus angle $\theta \in [0, 2\pi)$ was encoded over 32 evenly spaced preferred angles μ_j as

$$c_j(\theta) = \exp\{\kappa[\cos(\theta - \mu_j) - 1]\}, \qquad \kappa = 8. \tag{1}$$

The fixation input and output remained high until the response cue. Circular outputs were deliberately silent before response and then reported the remembered angle. Pre-cue and cue durations varied during training; evaluation used clean delays of 10, 20, 40, and 80 model steps. A separate delay-20 condition inserted a five-step irrelevant circular cue midway through the delay. Training sampled one clean and one distractor-present homogeneous batch exactly once per shuffled two-batch block. Distractor filtering was therefore a learned property of the evaluated networks rather than an out-of-distribution challenge.

Context-cued N-back task

The N-back task presented 20 sequential items drawn from six stimulus identities. Two context channels instructed either 0-back or 2-back behaviour. In 0-back, the network classified whether the current item matched a fixed target identity. In 2-back, it classified whether the current item matched the item presented two positions previously. Sequences contained six matches; the 2-back generator additionally controlled one-back lures. The first two items were excluded from scored outcomes. The output consisted of match and non-match logits.

Recurrent networks and checkpoint replication

Both task families used 64-unit `tanh` recurrent networks with $dt = 20$, $\tau = 100$, and $\alpha = dt/\tau = 0.2$. The native update was

$$z_t = W_{\text{in}} x_t + b_{\text{in}} + W_{\text{rec}} h_{t-1} + b_{\text{rec}}, \tag{2}$$
$$h_t = \tanh[(1-\alpha)h_{t-1} + \alpha z_t]\,. \tag{3}$$

The circular model had 33 inputs and outputs: 32 tuned channels plus fixation. The N-back model had six stimulus inputs, two rule-context inputs, and two outputs. Circular networks were trained for 4,000 Adam updates (learning rate 10^{-3}, batch size 64) with weighted fixation and response loss on a balanced clean/distractor mixture. Training used input noise SD 0.01, recurrent noise SD 0.05, and gradient clipping at 1.0; all learned weights were then frozen before perturbation. Six circular seeds were trained; five met preconfigured competence limits (clean mean error $\leq 10°$, distractor mean error $\leq 15°$, fixation accuracy ≥ 0.90) and were retained (seeds 20260731, 20260732, 20260733, 20260735, and 20260736). Seed 20260734 exceeded those limits (clean 15.75°, distractor 17.46°) and was excluded before the perturbation screen. The ten N-back checkpoints (seeds 20260912–20260921) were drawn from a competence-screened pool trained with staged 0-back then 2-back curriculum under Adam (learning rate 10^{-3}). Final N-back retention required accuracy ≥ 0.95, discriminability (hit rate minus false-alarm rate) ≥ 0.90, and 2-back lure accuracy ≥ 0.90 on both loads; stage-1 0-back acquisition used stricter gates (accuracy ≥ 0.98, discriminability ≥ 0.95) before 2-back training. A checkpoint is one independently trained network with frozen weights, identified by its random seed. Independently trained checkpoint, not trial or technical replicate, was the replication unit.

These gates are operational competence thresholds, not fitted human effect sizes. Chance circular error is of order 90°, so clean $\leq 10°$ requires unambiguous angle memory while remaining softer than the few-degree errors typical of retained seeds; distractor $\leq 15°$ allows a modest filtering cost without accepting near-failure. Fixation ≥ 0.90 keeps the network in a reliable go regime so latency contrasts are not confounded by fixation collapse. N-back gates likewise demand near-ceiling fixed-target performance, clearly above-chance updating, and lure resistance, so load contrasts reflect selective updating cost rather than a weak baseline.

Candidate perturbations

We evaluated seven single-operator families against each checkpoint's native baseline (Table I). Operators are competing computational interventions motivated by, but not equated with, psychedelic or working-memory mechanisms. Because the native update applies tanh after mixing drive with carried state, no operator is an exact biological response-gain map; each is an explicit RNN translation. Unless noted, every non-neutral operator was applied at every timestep of the full frozen forward pass for the entire trial or sequence; learned weights were never updated. The only timed exception is P3b, which is confined to the irrelevant circular-cue window. Write $z_t^{\mathrm{w}} = W_{\mathrm{in}}x_t + W_{\mathrm{rec}}h_{t-1}$ for the weight-derived drive. Symbols are shared across perturbations: x_t is input, h_t hidden state, and z_t pre-activation drive at time t; W_{in} and W_{rec} are input and recurrent weights; b_{in} and b_{rec} are biases; and $\alpha = dt/\tau$ is the native leak factor. In P3a/P3b, x_t^{stim} and x_t^{ctrl} denote stimulus versus fixation/rule-control channels, with matching input submatrices $W_{\mathrm{in}}^{\mathrm{stim}}$ and $W_{\mathrm{in}}^{\mathrm{ctrl}}$. Scalar g denotes uniform gain, vector $\mathbf{g}$ fixed unit-wise gain (P2), $\varepsilon_t \sim \mathcal{N}(0, I)$ Gaussian noise scaled by σ (P5), p persistence scale (P6), and s timescale factor with derived α' (P7). Unless noted, learned biases stay outside multiplicative gains.

P1 Synaptic-drive gain

Herzog et al. operationalised 5-HT2A effects as response gain on synaptic current inside a nonlinearity [1]; REBUS likewise links altered precision to postsynaptic gain [2]. In this CTRNN the closest available analogue multiplies weight-derived drive while leaving biases unscaled:

$$h_t = \tanh\left[(1-\alpha)h_{t-1} + \alpha\left(g z_t^{\mathrm{w}} + b_{\mathrm{in}} + b_{\mathrm{rec}}\right)\right]. \quad (4)$$

This is synaptic-drive gain, not an exact F–I slope change and not a pure carried-state manipulation.

P2 Heterogeneous drive gain

Receptor-density dependence in Herzog et al. implies spatially nonuniform gain rather than one global scalar [1]. We therefore replaced g by a fixed positive unit-wise vector $\mathbf{g}$ drawn from a log-normal distribution and normalised to population mean one:

$$h_t = \tanh\left[(1-\alpha)h_{t-1} + \alpha\left(\mathbf{g} \odot z_t^{\mathrm{w}} + b_{\mathrm{in}} + b_{\mathrm{rec}}\right)\right]. \quad (5)$$

Three fixed vectors were treated as technical replicates and averaged within checkpoint; they are not independent trained seeds.

P3a Sensory-input gain

REBUS predicts increased influence of afferent or bottom-up signals when high-level precision is relaxed [2]. As a task-level translation, only stimulus-channel input weights were scaled, leaving fixation or rule-context channels unchanged:

$$\begin{aligned} z_t^{\mathrm{in}} &= g\, W_{\mathrm{in}}^{\mathrm{stim}} x_t^{\mathrm{stim}} + W_{\mathrm{in}}^{\mathrm{ctrl}} x_t^{\mathrm{ctrl}} + b_{\mathrm{in}}, \\ h_t &= \tanh\left[(1-\alpha)h_{t-1} + \alpha(z_t^{\mathrm{in}} + W_{\mathrm{rec}}h_{t-1} + b_{\mathrm{rec}})\right]. \end{aligned} \quad (6)$$

This tests sensory sensitivity, not receptor-mapped response gain.

P3b Distractor-window gain

Carter et al. reported impaired multiple-object tracking with comparatively spared spatial working-memory span and proposed weakened filtering of irrelevant information [3]. P3b applies Eq. (6) only during the irrelevant circular cue and is therefore a circular-task manipulation check rather than a complete cross-task mechanism candidate.

P4 Recurrent gain

Recurrent excitation is a canonical substrate of delay-period maintenance [4]. To separate recurrent weighting from sensory drive we scaled only the recurrent weight contribution:

$$\begin{aligned} z_t^{(4)} &= W_{\mathrm{in}}x_t + b_{\mathrm{in}} + gW_{\mathrm{rec}}h_{t-1} + b_{\mathrm{rec}}, \\ h_t &= \tanh\left[(1-\alpha)h_{t-1} + \alpha z_t^{(4)}\right]. \end{aligned} \quad (7)$$

Recurrent bias remained outside the gain. P4 is a mechanistic contrast within the trained RNN, not a direct REBUS or 5-HT2A claim.

P5 Gaussian state noise

Independent Gaussian noise is retained as a generic disruption control against which selective signatures must later be shown to exceed nonspecific damage. Noise would be added to the pre-activation,

$$h_t = \tanh[(1-\alpha)h_{t-1} + \alpha z_t + \sigma\varepsilon_t], \quad \varepsilon_t \sim \mathcal{N}(0, I), \quad (8)$$

but P5 was not evaluated in this candidate-only screen.

P6 State persistence

Trained working-memory RNNs can store information along slow latent manifolds [5], while neurophysiological evidence links NMDA-dependent persistent firing to working-memory maintenance [4]. P6 asks whether weakening retention of the carried state, without rescaling incoming drive, can produce selective behavioural costs. The update is asymmetric:

$$h_t = \tanh[p(1-\alpha)h_{t-1} + \alpha z_t], \quad (9)$$

where p is the persistence scale. This is an operational RNN intervention, not a pre-specified biological mapping from psilocybin to reduced persistence.

P7 Conserved effective time constant

To test whether P6 effects are merely those of any timescale change, P7 rescales the leaky-integrator coefficients while preserving their sum. With timescale factor s,

$$\alpha' = \min(\alpha/s, 1), \qquad h_t = \tanh[(1-\alpha')h_{t-1} + \alpha' z_t]\,. \tag{10}$$

Thus $s > 1$ slows the effective timescale and $s < 1$ speeds it. Within the evaluated grid, $\alpha/s < 1$, so the minimum was inactive. P7 is a mechanistic control for conserved timescale change, not a psychedelic parameter claim.

Evaluation and behavioural signature

The evaluated contrasts were qualitative model analogues of selected acute human findings rather than fitted human effect sizes. A stronger acute effect on response time than accuracy motivated slowing with relative preservation [6]; selective 2-back impairment relative to 0-back motivated the N-back load contrast [7]; and impaired multiple-object tracking with comparatively spared spatial span motivated the filtering contrast [3]. Delay selectivity was retained as supporting mechanistic evidence rather than a scored human-signature requirement.

Each circular checkpoint, operator, strength, condition, and delay used 1,024 trials. Each N-back checkpoint, operator, strength, and load used 1,024 20-item sequences. Native and perturbed evaluations reused identical generated task samples within checkpoint and cell.

Circular response accuracy was mean absolute angular error. Post-cue settling was the first response step at which angular error was below 15° and population-vector amplitude exceeded 50% of the native reference amplitude. Trials that did not settle were capped at the 25-step response duration, yielding restricted mean settling time. Settling was interpreted only when fixation accuracy was at least 0.90 and at least 80% of trials settled. The 15° settling threshold is a coarse "pointing near the target" latency criterion, distinct from the continuous accuracy metric; the half-native amplitude rule excludes weak or collapsed population bumps; and the 80% settled floor marks latency invalid when most trials never settle. "Slowing with preservation" required a positive settling change at clean delay 20 while proportional angular-error impairment remained at or below 0.20. That 20% relative-error ceiling encodes selective slowing without large accuracy loss: small clean costs remain allowed, but wholesale damage fails the preservation half of the signature.

For circular condition c, proportional impairment was

$$I_c = \frac{E_{\text{perturbed},c} - E_{\text{native},c}}{E_{\text{native},c}}, \tag{11}$$

where E is mean angular error. Delay selectivity was $I_{\text{delay 80}} - I_{\text{delay 10}}$; distractor selectivity was $I_{\text{distractor 20}} - I_{\text{clean 20}}$.

N-back discriminability was $D = H - F$, where H and F are hit and false-alarm rates. Condition-normalised impairment was

$$J_c = \frac{D_{\text{native},c} - D_{\text{perturbed},c}}{D_{\text{native},c}}, \tag{12}$$

and load selectivity was $J_{\text{2back}} - J_{\text{0back}}$. Positive values indicate disproportionate high-load impairment.

The analysis was descriptive. For each operator-strength setting we reported the checkpoint mean, SD, and a student t 95% interval across seeds, together with the fraction of seeds in the predicted direction. A setting met the complete descriptive pattern when the mean slowing-with-preservation, distractor, and N-back load contrasts were positive and at least a simple majority of checkpoints agreed on each (three of five circular; six of ten N-back); delay selectivity was supporting evidence only. That majority rule is implemented in the summary code; it was not restated as a numeric threshold in the 1,024-trial preregistration text. Cross-operator comparisons are uncontrolled for overall perturbation magnitude, so a near-miss may fail because of strength selection as well as mechanism. No participant-level inference, confirmatory p values, or multiple-comparison claims were made.

Persistence 0.95 had been nominated by an earlier three-checkpoint exploratory pilot. The present analysis is therefore a retest on newly trained circular checkpoints and an expansion of the partially overlapping N-back evaluation, not a fresh blind screen. Three of the ten N-back checkpoints (seeds 20260912–20260914) contributed to the nomination pilot.

RESULTS

One descriptive profile met the complete majority pattern

Across the 23 non-neutral shared operator-strength profiles evaluated on the distractor-trained circular family and the ten N-back checkpoints, only persistence 0.95 met the complete descriptive majority pattern (Fig. 1). It showed slowing with preservation, long-delay selectivity, and distractor selectivity in four of five circular checkpoints, together with load selectivity in all ten N-back checkpoints. Mean circular delay selectivity was 0.089 and mean distractor selectivity 0.015; N-back load selectivity was 0.250. Neutral settings returned zero contrast. Other perturbations often reproduced one or two components, but not the full combination.

TABLE I. **Candidate perturbations and fixed evaluation grids.** Neutral settings reproduce the native forward pass. Distractor-window gain was defined only for the circular distractor condition; all other operators except P5 were evaluated in both task families. P5 is listed as the planned generic control but was not part of this screen.

ID	Perturbation	Computational intervention	Strengths
P1	Synaptic-drive gain	Scalar gain on input and recurrent weight drive	0.90, 0.95, 1.00, 1.05, 1.10
P2	Heterogeneous drive gain	Fixed mean-one unit-wise log-normal gain	log-SD 0.00, 0.10, 0.20, 0.30
P3a	Sensory-input gain	Gain on stimulus-channel input contribution	0.80, 0.90, 1.00, 1.10, 1.20
P3b	Distractor-window gain	P3a restricted to irrelevant cue window	1.00, 1.10, 1.25, 1.50
P4	Recurrent gain	Scalar gain on recurrent weight contribution	0.90, 0.95, 1.00, 1.05, 1.10
P5	Gaussian state noise	Additive pre-activation noise (not screened)	not evaluated here
P6	State persistence	Scale p on carried-state coefficient	0.90, 0.95, 1.00, 1.05, 1.10
P7	Effective time constant	Conserved update with $\alpha' = \min(\alpha/s, 1)$	0.80, 0.90, 1.00, 1.10, 1.25

A 5% persistence reduction preserved ordinary accuracy while selectively affecting demanding conditions

At $p = 0.95$, clean delay-20 proportional angular-error change was 0.049 ± 0.092 across circular checkpoints (95% t-interval $[-0.065, 0.164]$) and remained below 0.20 in 5/5 seeds. Restricted mean settling increased by 0.104 ± 0.167 steps ($[-0.103, 0.312]$) and was positive in 4/5 seeds, with a range of -0.013 to 0.393 steps (Table II). The circular settling, delay, and distractor intervals all include zero, so those components are directional majority tendencies rather than robust across-checkpoint effects. By contrast, N-back load selectivity was 0.250 ± 0.077 ($[0.195, 0.305]$), positive in 10/10 checkpoints, and ranged from 0.110 to 0.363. Long-minus-short delay selectivity was 0.089 ± 0.075 ($[-0.004, 0.181]$) and distractor-minus-clean selectivity was 0.015 ± 0.044 ($[-0.040, 0.071]$), each positive in 4/5 checkpoints. Thus the leading perturbation retained the predicted average ordering, but only the N-back contrast clearly excluded zero under the seed-level t-interval.

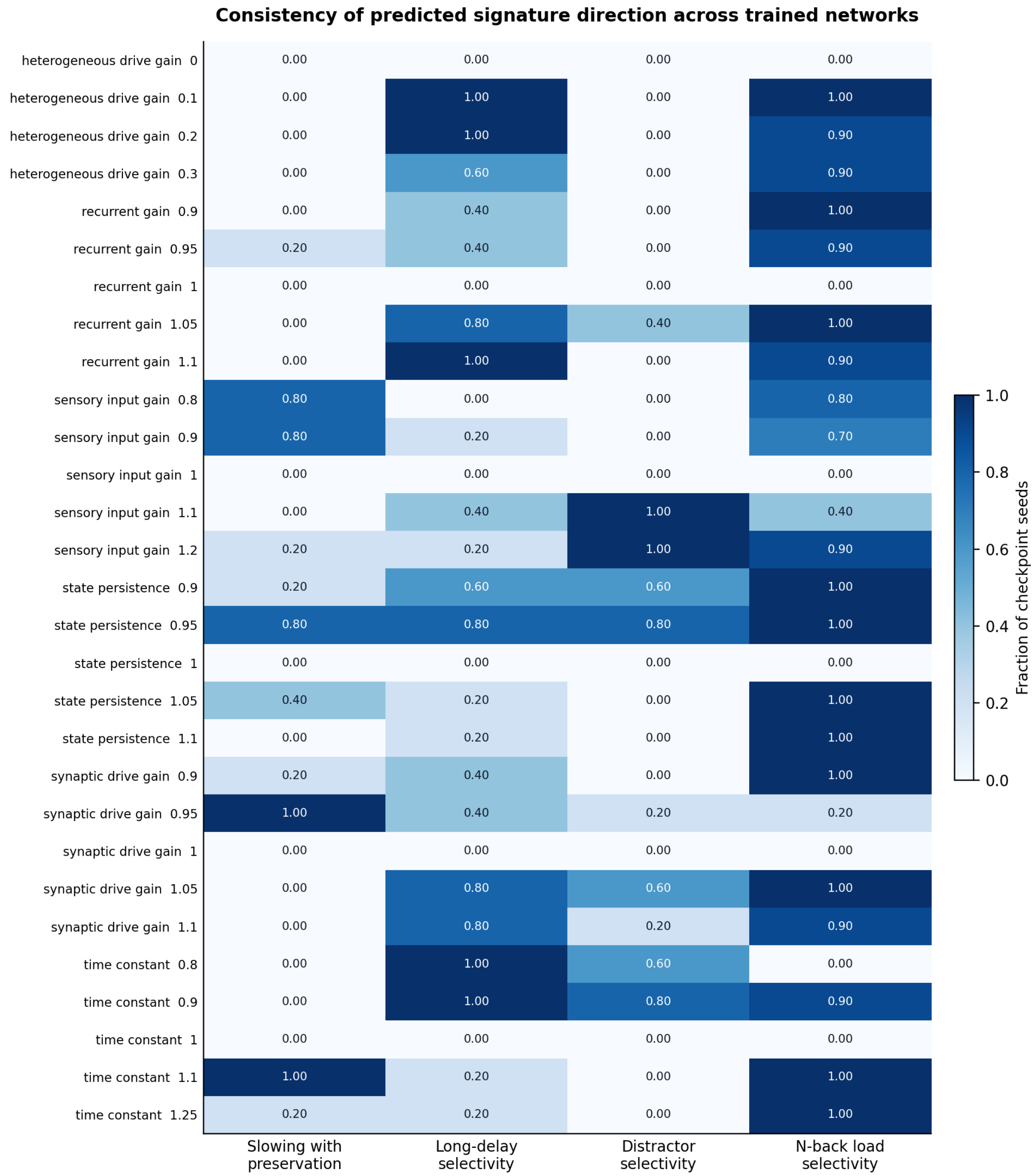
Persistence effects varied non-monotonically with perturbation strength

Reducing persistence further to 0.90 enlarged mean settling, delay, and N-back load effects (Fig. 2). However, slowing with preservation passed only 1/5 circular checkpoints, so the stronger intervention no longer expressed the selective accuracy-preserving pattern. Increasing persistence above one continued to impair N-back preferentially but reversed the circular delay and distractor contrasts. The joint signature was therefore specific to a weak reduction around 0.95 rather than monotonic damage at any non-neutral strength.

Alternative operators reproduced partial signatures

Sensory-input gain 1.20 produced distractor selectivity in 5/5 circular checkpoints and load selectivity in 9/10 N-back checkpoints, but mean settling became slightly faster (-0.010 ± 0.023 steps; positive in only 1/5) and delay selectivity was positive in only 1/5. Effective time-constant scale 1.10 slowed settling with preservation in 5/5 (0.605 ± 0.176 steps) and showed load selectivity in 10/10, but distractor selectivity was negative in every circular checkpoint. Conversely, time-constant scale 0.90 showed delay selectivity in 5/5 and distractor selectivity in 4/5 but accelerated settling in every circular checkpoint (-0.681 ± 0.170 steps). Synaptic-drive gain 1.05 showed delay selectivity in 4/5, distractor selectivity in 3/5, and load selectivity in 10/10, but also accelerated settling in 5/5 (-0.496 ± 0.240 steps). All of these near-miss clean delay-20 settling cells were latency-valid; uniqueness was therefore not an artefact of invalid latency scoring. These near-misses constrain the result: matching one difficult condition was insufficient to qualify as the complete pattern, and the discrimination against the conserved time-constant operator rested on settling direction and distractor selectivity rather than on large absolute settling magnitude.

The contrast between persistence and the conserved time-constant operator is mechanistically informative. Both reduced persistence ($p < 1$) and reduced timescale ($s < 1$) can weaken retention, but only persistence leaves the drive coefficient unchanged (Eq. 9). Reducing the conserved timescale ($s = 0.90$) accelerated post-cue settling while preserving delay and distractor selectivity; increasing it ($s = 1.10$) slowed settling but reversed distractor selectivity. The joint pattern therefore depended on the asymmetric carried-state intervention, not merely on any reduction in effective memory lifetime. Cross-operator comparisons remain uncontrolled for overall clean-task cost, so differences in perturbation magnitude could still contribute to some near-misses.



	Slowing with preservation	Long-delay selectivity	Distractor selectivity	N-back load selectivity
heterogeneous drive gain 0	0.00	0.00	0.00	0.00
heterogeneous drive gain 0.1	0.00	1.00	0.00	1.00
heterogeneous drive gain 0.2	0.00	1.00	0.00	0.90
heterogeneous drive gain 0.3	0.00	0.60	0.00	0.90
recurrent gain 0.9	0.00	0.40	0.00	1.00
recurrent gain 0.95	0.20	0.40	0.00	0.90
recurrent gain 1	0.00	0.00	0.00	0.00
recurrent gain 1.05	0.00	0.80	0.40	1.00
recurrent gain 1.1	0.00	1.00	0.00	0.90
sensory input gain 0.8	0.80	0.00	0.00	0.80
sensory input gain 0.9	0.80	0.20	0.00	0.70
sensory input gain 1	0.00	0.00	0.00	0.00
sensory input gain 1.1	0.00	0.40	1.00	0.40
sensory input gain 1.2	0.20	0.20	1.00	0.90
state persistence 0.9	0.20	0.60	0.60	1.00
state persistence 0.95	0.80	0.80	0.80	1.00
state persistence 1	0.00	0.00	0.00	0.00
state persistence 1.05	0.40	0.20	0.00	1.00
state persistence 1.1	0.00	0.20	0.00	1.00
synaptic drive gain 0.9	0.20	0.40	0.00	1.00
synaptic drive gain 0.95	1.00	0.40	0.20	0.20
synaptic drive gain 1	0.00	0.00	0.00	0.00
synaptic drive gain 1.05	0.00	0.80	0.60	1.00
synaptic drive gain 1.1	0.00	0.80	0.20	0.90
time constant 0.8	0.00	1.00	0.60	0.00
time constant 0.9	0.00	1.00	0.80	0.90
time constant 1	0.00	0.00	0.00	0.00
time constant 1.1	1.00	0.20	0.00	1.00
time constant 1.25	0.20	0.20	0.00	1.00

FIG. 1. **Cross-task signature screen.** Rows are every shared operator-strength setting evaluated in both task families; columns show the fraction of independent checkpoint seeds with the predicted signature direction. Circular fractions have denominator five and N-back fractions denominator ten. State persistence 0.95 was the only row to meet the complete majority pattern and additionally reached 4/5 agreement on each circular feature with complete N-back agreement. Values quantify directional consistency, not participant-level probability or statistical significance.

Native distractor competence and validity gates

All 945 circular result rows passed the fixation gate, all ten N-back native checkpoints passed competence, and all $p = 0.95$ clean delay-20 cells were latency-valid. Twelve rows elsewhere in the circular grid were latency-invalid (all heterogeneous-drive settings with low settled fraction); none of the near-miss settling comparisons above used those cells. Native distractor mean error ranged from 3.46° to 4.83°, and every native distractor cell had a settled fraction of 1.00. The distractor contrast therefore measured perturbation of a learned filtering behaviour rather than failure on an unfamiliar task. As a manipulation check, distractor-window gains 1.10, 1.25, and 1.50 increased distractor error by 3.7%, 9.8%, and 21.5% on average, respectively, with positive effects in

TABLE II. **State persistence 0.95.** Values are mean ± SD across five circular or ten N-back checkpoint seeds; intervals are student t 95% intervals across those seeds.

Outcome	Contrast	95% interval	Consistency
Delay-20 error	0.049 ± 0.092	$[-0.065, 0.164]$	below 0.20 in 5/5
Settling change (steps)	0.104 ± 0.167	$[-0.103, 0.312]$	4/5 positive
Delay selectivity	0.089 ± 0.075	$[-0.004, 0.181]$	4/5 positive
Distractor selectivity	0.015 ± 0.044	$[-0.040, 0.071]$	4/5 positive
N-back load selectivity	0.250 ± 0.077	$[0.195, 0.305]$	10/10 positive

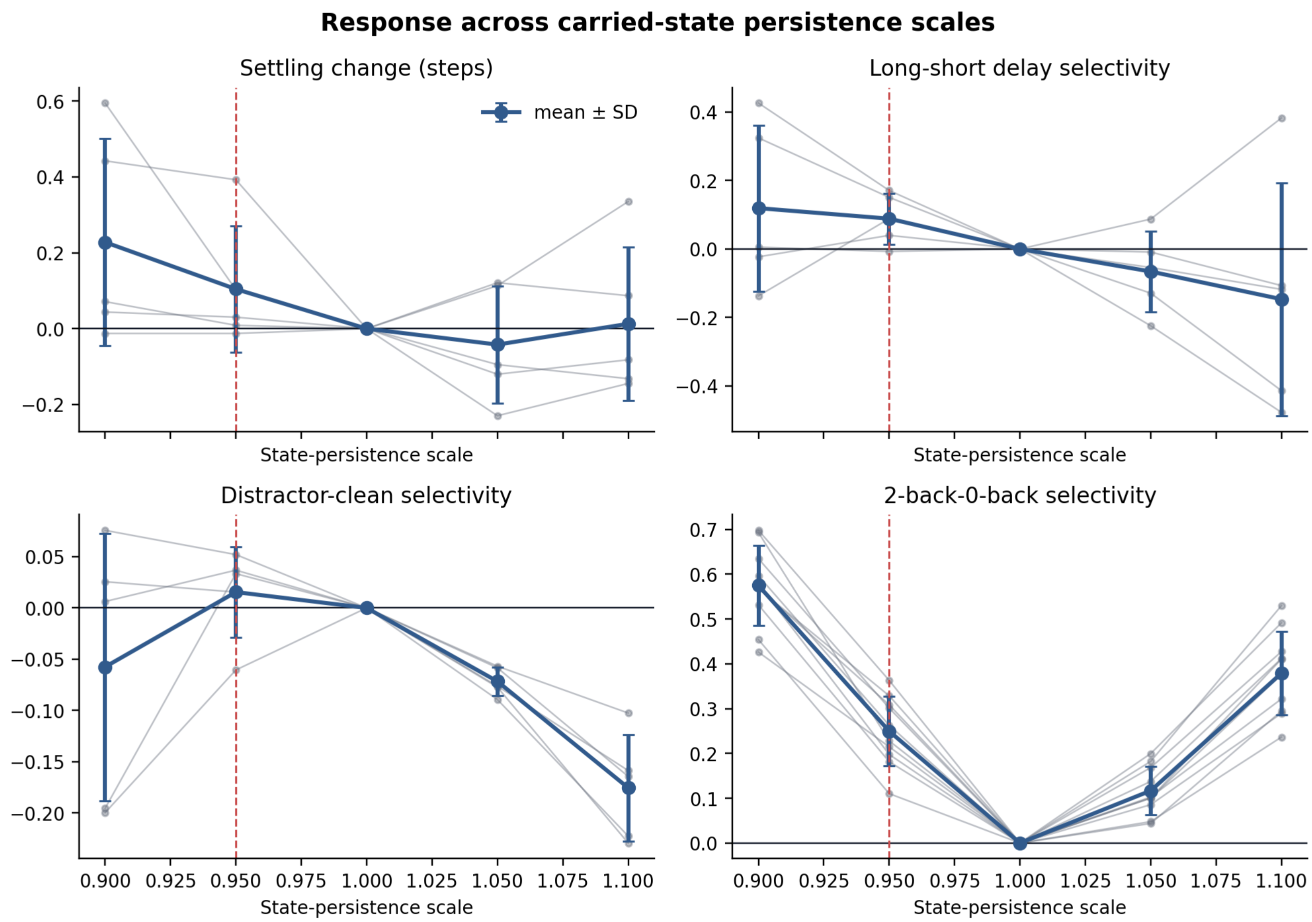


FIG. 2. **Response across carried-state persistence scales.** The four panels report, from top-left to bottom-right: change in restricted mean settling steps, long-minus-short delay selectivity ($I_{\text{delay }80} - I_{\text{delay }10}$), distractor-minus-clean selectivity ($I_{\text{distractor }20} - I_{\text{clean }20}$), and 2-back-minus-0-back load selectivity ($J_{\text{2back}} - J_{\text{0back}}$). Positive values indicate slower settling or larger impairment in the more demanding condition. Grey lines are independently trained checkpoint seeds; blue points and error bars are mean ± SD across seeds. The dashed red line marks the leading persistence scale 0.95 and the horizontal line marks no change from each checkpoint's native baseline. A stronger reduction increased several impairments but also violated relative preservation, whereas increased persistence reversed circular delay and distractor selectivity.

5/5 checkpoints. Persistence-related distractor selectivity nevertheless remained small and was negative in one checkpoint, so competent native filtering does not by itself make that contrast a strong human effect-size match.

This screen did not include a matched-cost Gaussian comparator. Specificity against generic disruption therefore remains an open next experiment rather than a claim supported here.

DISCUSSION

Reduced carried-state persistence was the only evaluated profile to meet the complete descriptive majority rule for the selected cross-task behavioural pattern against the native baseline. Scaling the carried-state coefficient by 0.95 left ordinary circular accuracy approximately intact while modestly slowing response settling, increasing vulnerability at longer delays and under distraction, and selectively impairing 2-back discriminability. The circular components had the predicted mean signs and majority directional agreement, but their seed-level intervals included zero; only the N-back load contrast was robust across checkpoints.

Equation 9 suggests a coherent model-level explanation. Each update retains slightly less of the preceding hidden state without simultaneously rescaling incoming drive. Information is therefore not erased at once; rather, its stability is weakened cumulatively. Short and simple conditions remain relatively intact, whereas longer maintenance, competing input, and repeated updating expose the loss. The conserved time-constant near-misses support this asymmetry reading: when carried-state and drive coefficients are rescaled together, the joint signature breaks. The present behavioural analyses do not directly demonstrate a changed attractor or slow manifold.

Importantly, "state persistence" is an operational property of this implemented recurrence, not a measured neurobiological parameter. Under the descriptive scoring rule, the result shows candidate-versus-baseline qualitative sufficiency: this intervention can generate the selected pattern. It does not show that psilocybin reduces neural persistence, that 5-HT2A activation maps to $p = 0.95$, or that the mechanism is unique. A matched generic-damage comparator remains the outstanding specificity test. Several alternative operators also reproduced individual components. Subsequent analyses should therefore examine hidden-state drift and stability and, separately, test specificity against non-neutral matched-cost controls.

Two limitations particularly constrain the present inference. First, post-cue settling is not literal reaction time because response onset was externally imposed; moreover, the mean change was substantially less than one model step and its across-seed interval included zero. Second, although the circular networks were competent under distraction, the persistence distractor contrast was only 0.015 on average, reversed in one checkpoint, and also had an interval including zero. The circular delay contrast was likewise small (0.089 ± 0.075). The circular task is also not the multiple-object tracking paradigm used by Carter et al. [3]; it tests a related filtering principle rather than reproducing that task. The descriptive grid contains multiple operator-strength profiles but no confirmatory inferential family. Checkpoint consistency demonstrates robustness across trained models, not population inference about human participants.

The immediate next experiment should compare persistence 0.95 with matched-cost Gaussian disruption on these circular and N-back families, then densify persistence scaling around 0.95 and quantify delay-period representational drift on independently banked task samples. That specificity analysis can retain the candidate-versus-baseline result reported here while testing whether the pattern exceeds generic damage.

LIMITATIONS AND NEXT STEPS FOR SUPERVISOR DISCUSSION

Key limitations to discuss. The central weakness is asymmetry across task families: N-back load selectivity was robust, but circular delay and distractor selectivity were small and their seed-level intervals included zero. The settling endpoint is also an imposed model proxy rather than biological reaction time, and the observed slowing was well below one model step on average. Mechanistic interpretation remains behaviour-first: this report does not yet show whether persistence scaling changes attractor geometry, manifold drift, or recovery dynamics in a way that uniquely distinguishes it from other weak perturbations. Finally, the current screen is candidate-versus-native only; without a matched-cost Gaussian comparator, specificity against generic disruption remains untested.

Immediate next steps. First, run the matched-cost Gaussian comparator on the same checkpoints and banked samples, with preregistered cost matching to separate selective signatures from generic damage. Second, densify persistence around 0.95 (for example 0.93–0.98) to test whether the joint pattern is locally stable or a narrow parameter point. Third, add targeted dynamics analyses on identical trials: delay-period drift, local Jacobian spectrum near delay states, and perturbation recovery trajectories after distractor input. Fourth, predefine a compact confirmatory analysis family for the shortlisted contrasts so the next report can move from descriptive directional consistency to controlled inferential claims.

CONCLUSION

A 5% reduction in carried-state persistence was the only tested RNN perturbation to reproduce the complete descriptive majority pattern on distractor-trained circular networks and competence-screened N-back networks: relative preservation with slower settling, greater long-delay and distractor costs, and selective 2-back impairment. The circular pattern held in four of five checkpoints and circular effect sizes were small, whereas N-back load selectivity was robust across all ten checkpoints. Reduced persistence remains a credible candidate for matched-cost specificity and mechanistic analysis, but the present result is an abstract descriptive sufficiency finding against the native baseline, not evidence for a biological mechanism of psilocybin.

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